

ALLEN IOQM Number Theory Marathon

Extracted Questions, Answers & Problem-Solving Tips - Lot 1 (Q1-Q5)

Video: <https://www.youtube.com/live/0j8W6Q8ID8A>

Session: Number Theory for IOQM | Live Marathon | ALLEN

Method: Question reconstruction from the available teaching discussion, followed by independent mathematical checks.

Accuracy note: Q2-Q5 are mathematically cross-checked. Q1 remains wording-sensitive: the currently available reconstruction and the stated answer do not match a plain right-associated power tower of 9s, so it is deliberately marked unverified.

Q1. Repeated-9 exponent expression (wording not yet verified)

Reconstructed question: Compute the last two digits of a 2020-level nested exponent expression involving repeated 9s.

Answer stated in the available reconstruction: 20 (unverified).

Important check: If the expression were simply the right-associated tower $9^{(9^{(9^{(...)})})}$, then a height of at least 2 gives a residue inconsistent with 20 modulo 100. Therefore some part of the original on-screen expression is missing from the transcript reconstruction.

Tip: For last-two-digit problems, immediately work modulo 100. For exponent towers, first find the multiplicative period/order of the base modulo 100, then reduce the exponent modulo that period. Confirm the exact placement of parentheses: exponentiation is right-associative.

Q2. Largest two-digit factor of a huge difference

Question: Find the largest positive two-digit factor of $3^{(2^{2011})} - 2^{(2^{2011})}$.

Answer: 97.

Key reasoning: Repeatedly use $x^{(2m)} - y^{(2m)} = (x^m - y^m)(x^m + y^m)$. Since 2^{2011} is a power of 2, the factorization contains $3^4 + 2^4 = 81 + 16 = 97$. The number is odd, so 98 cannot divide it. It is also not divisible by 3, so 99 cannot divide it. Hence 97 is the largest two-digit factor.

Tip: When the exponent itself is a power of 2, look for the chain $x-y$, $x+y$, x^2+y^2 , x^4+y^4 , ... before doing modular arithmetic. A large prime can appear as a small cyclotomic-type factor.

Q3. Factorials after deleting trailing zeros

Question: Richard writes $n!$ and $(n+1)!$ on a board and erases all trailing zeros from each, leaving integers a and b . If one of a and b is 4 times the other, find the last two digits of the sum of all possible n with $n < 1000$.

Answer: 14. The valid values are $n = 3, 24, 39, 249, 399$, whose sum is 714.

Key reasoning: Let $v_5(n+1)$ be the exponent of 5 in $n+1$. Passing from $n!$ to $(n+1)!$ introduces $n+1$ and exactly $v_5(n+1)$ new trailing zeros, so after stripping zeros, $b/a = (n+1)/10^{v_5(n+1)}$. Set this ratio equal to 4 or $1/4$. This gives $n+1 = 4, 40, 400$ or $n+1 = 25, 250$.

Tip: Never calculate the factorials. Compare the two zero-stripped factorials by a ratio. Trailing-zero questions are really questions about powers of 5, because factorials contain far more factors of 2 than factors of 5.

Q4. Preserving LCM(1,2,...,50)

Corrected question: Find the largest positive integer n such that $\text{lcm}(n, n+1, \dots, 50) = \text{lcm}(1, 2, \dots, 50)$.

Answer: 27.

Key reasoning: An LCM is determined by the largest required prime powers. In $\text{lcm}(1, \dots, 50)$, the critical powers include $2^5=32$, $3^3=27$, $5^2=25$ (also supplied by 50), and $7^2=49$, together with the primes up to 50. Starting at 27 still retains every required prime-power contribution. Starting at 28 loses the unique factor 3^3 , because no number from 28 through 50 is divisible by 27.

Tip: For interval-LCM problems, do not track every omitted integer. Track only the maximal prime powers p^k that define the LCM. Ask whether the shortened interval still contains a multiple carrying each required p^k .

Q5. Balance three integer factors of 7!

Question: Positive integers x, y, z satisfy $xyz = 7!$. Minimize $\max(x, y, z)$.

Answer: 20, achieved by $(x, y, z) = (14, 18, 20)$ in any order.

Key reasoning: $7! = 5040$ and the cube root of 5040 is about 17.1, so the optimum should be near 17-20. If all three factors were at most 19, each would have to be a divisor of 5040 not exceeding 19. The only pair products large enough to leave the third factor at most 19 are essentially $18 \times 16 = 288$ or $18 \times 15 = 270$, and neither leaves an integer factor at most 19. Thus 19 is impossible. The factorization $14 \times 18 \times 20 = 5040$ proves 20 is achievable.

Tip: In minimize-the-maximum factor problems, start from the cube root of the product. Then search for a balanced divisor triple near that scale; after finding a candidate, prove that one unit lower is impossible using divisibility and a narrow range of possible factors.

Lot 1 status: Q2-Q5 are worksheet-ready. Q1 should stay flagged until the exact slide or a fuller transcript segment is available.

Q3 candidate set was independently re-derived algebraically from the reconstructed statement. Q4 wording was corrected to largest n so that the answer 27 is mathematically consistent.